MNE 452 – Vehicle Dynamics **Exam 1** **Name: _____**

Brennan's grading summary:	
do NOT write in this box!	
Name every page:	_____/5
Prob1:	_____/5
Prob2:	_____/15
Prob3:	_____/15
Prob4:	_____/10
Prob5:	_____/10
Prob6:	_____/20
Prob7:	_____/20
Total:	_____ / 100

(5 pts) Put your name at the top of <u>every</u> page.

1. (5 pts) Short answer: Which driving situation is most likely to result in a side impac: front wheel lock up or rear wheel lock up?
   _____

2. (15 pts) In class we discussed how to solve a differential equation using Euler's method. A first-order system with input $u(t)$ and output $y(t)$ is given by:

$$\frac{dy}{dt} = f(y,t,u) = -2y(t) + u(t) \quad (*)$$

This has a transfer function of: $\dfrac{Y(s)}{U(s)} = \dfrac{1}{s+2}$

a) (3 points) What is a good time step to simulate this system?

$\Delta t =$_____ sec

b) (5 points) Applying Euler's method to this system what is the equation for $y(t + \Delta t)$, assuming the input is zero (e.g. $u(t) = 0$)?

$y(t + \Delta t) =$_____

c) (7 points) To show that you know how to use this formula, calculate $y(1)$ assuming a step size of 0.5 seconds, $u(t) = 0$, and an initial condition of $y(0) = 1$. Show your intermediate result and final result in the space below. (I'm looking for numbers, not equations.)

$y(0.5) =$_____

$y(1) =$_____

3. (15 pts) Imagine this piece of paper is a road. Your vehicle is oriented so that the tires are sitting on the paper, and the front of the vehicle is pointing to the top of the paper in a straight line,

   Part 1 (6 points): If you do the following rotations with SAE convention, where are X and Y axes pointing after each of the maneuvers, assuming each is just one step in a sequence?

   Your answer should include the words: top of page, bottom of page, right, left, out of page, into page

	X axis	Y axis
a) 90 degrees yaw,	_____	_____
b) Then from there, -90 degrees roll,	_____	_____
c) Then, from there, -90 degrees yaw,	_____	_____

   Part 2: (3 points): Which rotation will bring the position you ended up in c) BACK to the starting point?

   _____

   Part 3: (3 points) If you change the order of rotations from c) to b) then a) and use ISO instead of SAE for each of the steps, do you end up in the same position?

   _____ (yes or no)

   Part 4: (3 points) Regardless of your answer in Part 3, do coordinate rotations *generally* give you the same answer if you mix up the order of a rotation sequence?

   _____ (yes or no)

4. (10 pts) A car is being steered to the right at 20 degrees of road tire angle. Assume both situations below are steady state, and use SAE coordinate systems.

Answer the following questions by circling the correct answer.

<u>At very low speeds:</u>

The sideslip angle (in degrees) of the vehicle is between:

-180 and -90 or -90 to 0 or 0 to 90 or 90 to 180

Or, can't answer because it depends on _____

The yawrate is going to be

positive negative Or, can't answer because it depends on _____

<u>Same situation, except at very high speeds</u>:

The sideslip angle (in degrees) of the vehicle is between:

-180 and -90 or -90 to 0 or 0 to 90 or 90 to 180

Or, can't answer because it depends on _____

The yawrate is going to be

positive negative Or, can't answer because it depends on _____

5. (10 points) The tie rod broke on a vehicle and suddenly the front tires of a vehicle are pointing in different directions. Determine the following turn centers, and label each as A, B, or C in the diagram below:

   A: The turn center if the driver brakes so hard that the rear tires were off the ground
   B: The turn center if the left tires were off the ground
   C: The turn center if the right tires were off the ground

6. (20 points) For a passenger car of track width 1.00 m and wheelbase 2.00 m (e.g.. distance from front axle to rear axle), what is the EXACT steering angles required for each of the front wheels in order for a vehicle to navigate a turn of 10 m radius at very low speeds? (assume the radius is measured to center of the rear axle). Write your answer in trigonometric functions (no need to calculate a numerical answer). Do NOT use small-angle approximations (-5 points if you do!)

$\delta_{f.inside} =$ _____ $radians$

$\delta_{f,outside} =$ _____ $radians$

For this vehicle, what is the sideslip angle of the bicycle model representation assuming the CG is in the middle of the vehicle, again assuming low speed? You can use either the exact solution or the small angle approximation, whichever you prefer… but clearly explain which one you are assuming!

Assuming that _____, my calculation of sideslip:

Gives $\beta =$ _____ $radians$

If this vehicle were the same track width but much longer (like a bus), would the sideslip angle be larger or smaller? _____ . Why is this result important for driving buses around the Penn State campus?

_____

7. (20 points) For a vehicle with track width 1.00 m and wheelbase 2.00 m, it has a weight of 1700 kg, a front/rear weight distribution of 60/40, and equivalent cornering stiffness of 100,000 N/rad for both front tires together, and 100,000 N/rad for both the rear tires together.

   a) Is this vehicle understeer or oversteer? _____ Explain your answer using the following equation)

   $$\delta = \frac{L}{R} + \left(\frac{b}{C_{\alpha f}} - \frac{a}{C_{\alpha r}} \right) \frac{mU^2}{LR}$$

   _____

   b) What is the critical or characteristic speed of the vehicle? (feel free to set this problem up if math gets too hard… you'll get full credit if you set it up correctly!)

   Critical OR Characteristic speed is: (circle the one you are solving for!)
   _____ m/s

   c) Sketch the relationship between front steering angle on the y-axis and speed on the x axis for this vehicle, specifically showing the appropriate curve (line, exponential, parabola, cubic, etc) for whether this is an oversteer, understeer, or neutral steer vehicle. Clearly label the following:

   Critical speed or Characteristic speed

   Ackerman angle

   d) If you wanted to take this vehicle road-racing, and could change ONLY the rear tires, what cornering stiffness would you choose, and why?

   $C_{\alpha r} =$ _____ N/rad

   Why? _____